

Python Application Programming

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Nested Tuple

```
]: # Create a nest tuple  
  
NestedT =(1, 2, ("pop", "rock") ,(3,4),("disco",(1,2)))
```

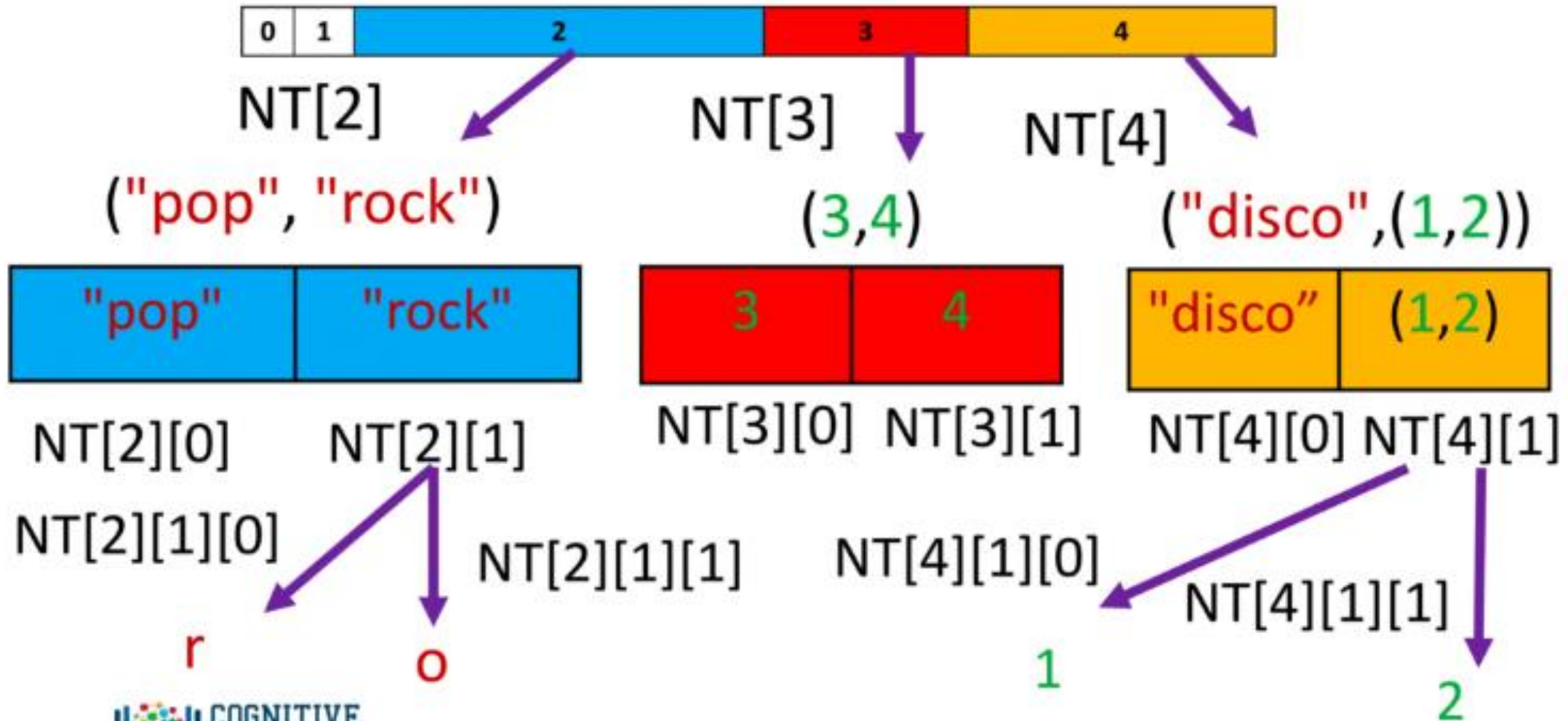
Each element in the tuple, including other tuples, can be obtained via an index as shown in the figure:

NT =(1, 2, ("pop", "rock") ,(3,4),("disco",(1,2)))



```
]: # Print element on each index  
  
print("Element 0 of Tuple: ", NestedT[0])  
print("Element 1 of Tuple: ", NestedT[1])  
print("Element 2 of Tuple: ", NestedT[2])  
print("Element 3 of Tuple: ", NestedT[3])  
print("Element 4 of Tuple: ", NestedT[4])  
  
Element 0 of Tuple: 1  
Element 1 of Tuple: 2  
Element 2 of Tuple: ('pop', 'rock')  
Element 3 of Tuple: (3, 4)  
Element 4 of Tuple: ('disco', (1, 2))
```

NT=(1, 2, ("pop", "rock"), (3,4), ("disco", (1,2)))



Dictionary

- ▶ A dictionary consists of keys and values. It is helpful to compare a dictionary to a list. Instead of being indexed numerically like a list, dictionaries have keys. These keys are the keys that are used to access values within a dictionary.
- ▶ The best example of a dictionary can be accessing person's details using the **social security number**.
Here the social security number which is a unique number will be the **key** and the details of the people will be the **values** associated with it.

Dictionaries

In list we have indexes but in dictionaries we have keys



Denoted by {}



Can be duplicates, immutable, mutable



Delete : `del(dict[keyvalue])`

Difference Between List and Dictionary

List

Index

0	Element 1
1	Element 2
2	Element 3
3	Element 4
.....	

Element

Dictionary

Key: is a index by label

Key 1	Value 1
Key 1	Value 2
Key 2	Value 3
Key 3	Value 4
.....	

Element/Values

Operations in Dictionaries

```
|: # Append value with key into dictionary

release_year_dict['Graduation'] = '2007'
release_year_dict
```

```
|: {'Thriller': '1982',
    'Back in Black': '1980',
    'The Dark Side of the Moon': '1973',
    'The Bodyguard': '1992',
    'Bat Out of Hell': '1977',
    'Their Greatest Hits (1971-1975)': '1976',
    'Saturday Night Fever': '1977',
    'Rumours': '1977',
    'Graduation': '2007'}
```

We can delete an entry:

```
|: # Delete entries by key

del(release_year_dict['Thriller'])
del(release_year_dict['Graduation'])
release_year_dict
```

```
|: {'Back in Black': '1980',
    'The Dark Side of the Moon': '1973',
    'The Bodyguard': '1992',
    'Bat Out of Hell': '1977',
    'Their Greatest Hits (1971-1975)': '1976',
    'Saturday Night Fever': '1977',
    'Rumours': '1977'}
```